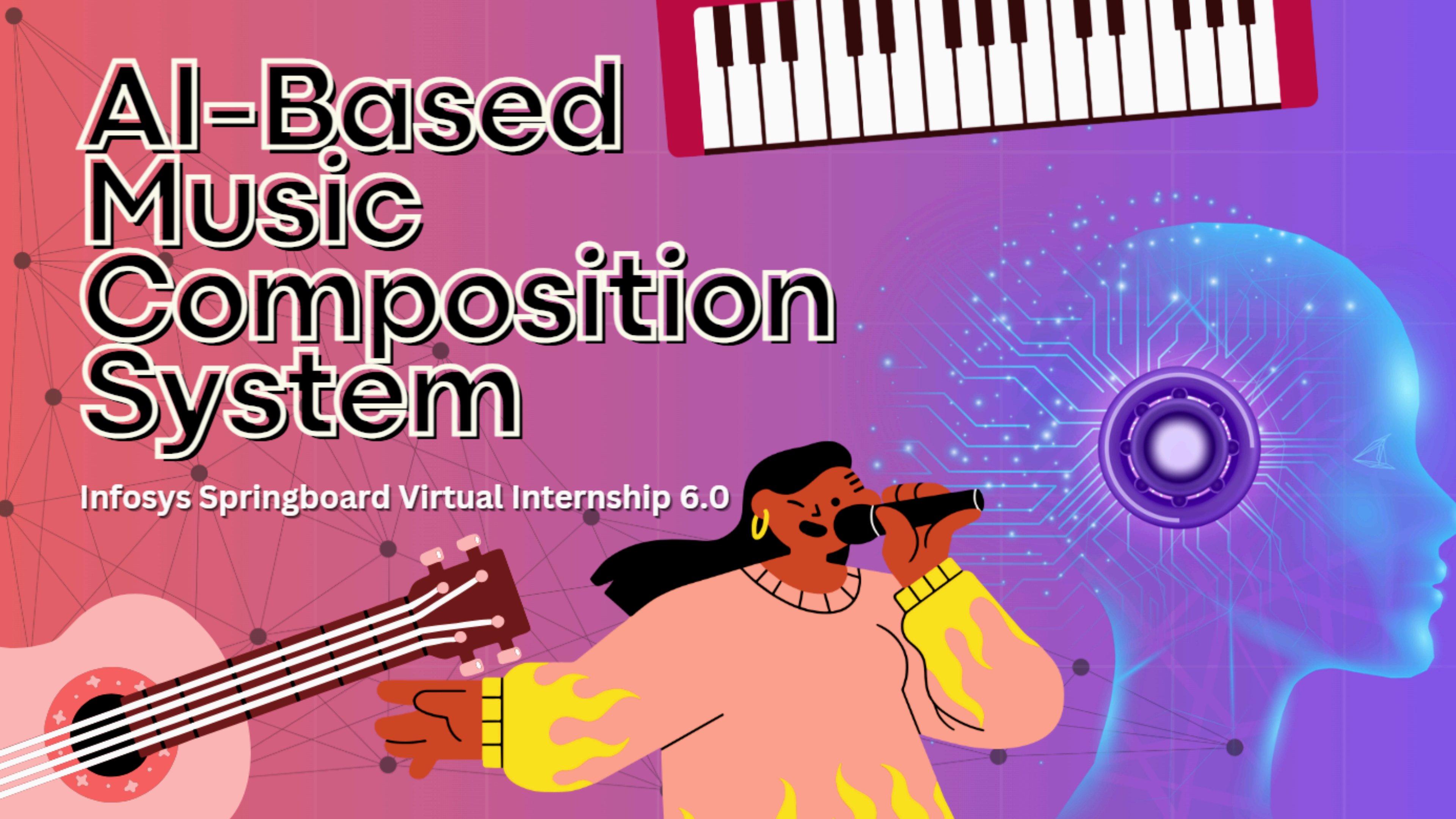


AI-Based Music Composition System

Infosys Springboard Virtual Internship 6.0



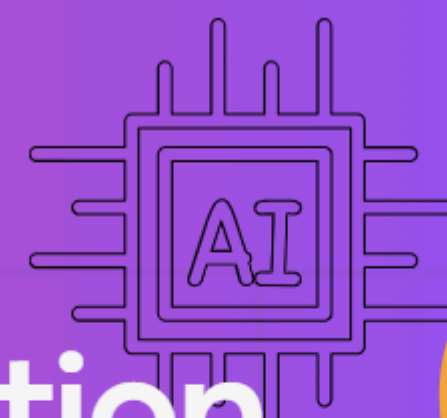
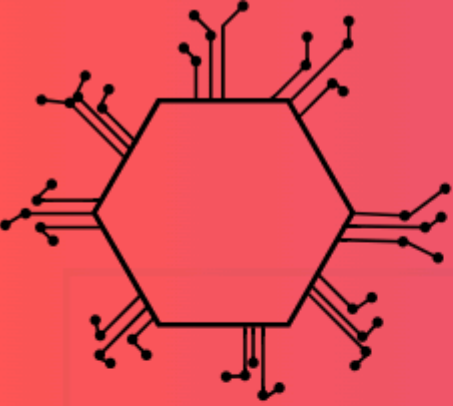


Introduction

This AI-powered music generation system that composes unique instrumental tracks based on user input.

- We focused on combining AI, ML, and web development to automate a complex creative process. Our goal was to let anyone create personalized, emotion-driven music clips effortlessly. The system enables fast content creation, background scoring, and experimental sound design.





The Need for AI-Assisted Composition

The Challenge of Manual Composition:

Creating original music traditionally requires expertise, time, and costly production tools, making creative audio generation hard to access

Lack of Emotional Personalization:

Most automated music generators create generic outputs and struggle to produce personalized tracks that match a user's exact mood, genre, or context

The Solution: AI Music Composer:

We built a responsive web-based AI Music Composer that uses a combined ML architecture to instantly generate customized music clips from simple text or category inputs, powered by Flask with YAMNet and MusicGen

Project Architecture and Flow



Core Technology Stack



Frontend:

React, Tailwind CSS, Framer Motion — Sleek, animated, clean interface.



Backend & Framework:

The core application logic was managed using the light-weight Flask framework in Python.



Key Libraries:

Librosa, NumPy, TensorFlow, and HuggingFace Transformers were crucial for audio processing and model interaction.



AI Models:

MusicGen for music generation, GPT-2 for lyrics, and YAMNet + GTZAN for genre classification



Overcoming Development Hurdles

Challenges	Description
High Latency in Generation	Optimized MusicGen inference and capped clips at 30 seconds for faster performance.
Managing Large Model Initialization	Large models like MusicGen and GPT-2 caused load delays, so we preloaded them locally and kept them in memory to avoid repeated loading.
UI Responsiveness	Browser playback and layout issues were fixed through CSS tweaks, responsive design testing, and improved event handling.

The Roadmap Ahead: Future Scope



Cloud-Based Hosting:

Move the Flask app and AI models to a scalable cloud platform like AWS or GCP for public access



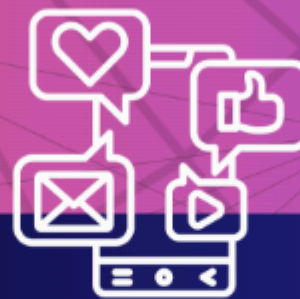
Real-Time Editing Tools:

Add in-browser tools for mixing, trimming, looping, and applying basic effects to generated tracks.



Multi-User Collaboration:

Enable multi-user accounts for co-composing and sharing libraries, building a community of AI-assisted music creators.



Streaming Service Integration:

Explore integration with APIs like Spotify or SoundCloud for seamless direct uploading and sharing of generated content.



Project UI Showcase

AI-Based Music Composer

Home

Compose

Library

Playlist



Logout

Compose AI Music Effortlessly

Generate tracks from text prompts, manage your library, and share your creations.

Generate Music

Explore Library



AI Music Generation

Create unique songs tailored to your mood and preferences.



Smart Lyrics

Generate meaningful lyrics that match your musical vision.



Personal Library

Save and organize your creations in your music library.

Compose

Describe the music you want to generate.

Moody cinematic strings with slow build and percussion

Generating...

Generate Lyrics

Random prompt

Upload audio for genre prediction

Select a local audio file. It will appear in Library and Playlist under its predicted genre.

Choose file

Uploaded successfully Predicted genre: **Reggae**



Generation in progress...

Moody cinematic strings with slow build and percussion

Thank You

